Supplementary Information

Title: Auditing PPI Graph Augmentation for Missense Variant Pathogenicity Prediction under Topology, Connectivity and Homology Constraints

Notes on scope

This Supplementary Information provides reconstruction details, split statistics, model comparisons, uncertainty estimates, topology-stratified summaries and negative-control results. Full figure source data, topology-stratified metrics, corrected-sample statistics and fusion-control summaries are included in the prepared compact source-data archive.

Supplementary Methods

Supplementary Methods 1. Dataset reconstruction and source files

We reconstructed a missense variant pathogenicity dataset from the ClinVar `variant_summary.txt.gz` table obtained from the official NCBI ClinVar FTP `tab_delimited` directory. We used the ClinVar variant-summary release dated 31 March 2024 and retained checksums for the exact input file used in the reconstruction. We retained germline GRCh38 single-nucleotide variant records and extracted protein-level HGVS annotations. Missense substitutions were identified using a strict three-letter amino-acid substitution pattern.

Pathogenic, Likely pathogenic and Pathogenic/Likely pathogenic records were grouped as pathogenic. Benign, Likely benign and Benign/Likely benign records were grouped as benign. Supported review status was defined as the exact ClinVar `ReviewStatus` strings `criteria provided, single submitter`, `criteria provided, multiple submitters, no conflicts`, `reviewed by expert panel`, and `practice guideline`; records outside this whitelist were excluded. Conflicting VariationIDs were removed, and duplicate VariationIDs were collapsed after conflict removal. The final mapped dataset contained 177,213 filtered, non-conflicting ClinVar missense variants with criteria-supported review status, including 127,953 benign and 49,260 pathogenic variants.

ClinVar gene symbols were mapped to STRING human protein nodes using upper-cased gene symbols and the STRING protein alias table. Only genes that mapped uniquely to a STRING protein present in the STRING links graph were retained. Mapping was performed at the protein-identifier level; isoform-specific effects and transcript-level ambiguity were not modeled separately. The reconstructed ClinVar-STRING dataset covered 14,123 unique variant-labelled STRING proteins embedded in a larger STRING node universe containing 19,622 proteins. The final variant tables retained genomic coordinates, reference and alternate VCF alleles, VariationID, binary label, gene symbol and mapped STRING protein identifier.

CADD and AlphaMissense scores were matched to ClinVar variants by GRCh38 genomic coordinates and reference/alternate alleles. CADD GRCh38 v1.7 whole-genome SNV scores were matched for 177,205 of 177,213 variants, leaving eight missing CADD rows. AlphaMissense hg38 scores were matched for 171,753 variants, leaving 5,460 missing AlphaMissense rows. CADD-only and AlphaMissense-only baselines used their corresponding matched scalar scores. CADD+AlphaMissense models used CADD PHRED, AlphaMissense pathogenicity and an AlphaMissense-missing indicator. Raw-ESM fusion concatenated these scalar features with the 1,280-dimensional raw ESM protein representation, whereas graph-fusion models concatenated the same scalar features with the 1,280-dimensional PPI-GCN representation. Follow-up graph-control fusions used the same scalar CADD+AlphaMissense features concatenated with real PPI-GCN, direct train-test edge-ablated PPI-GCN, or topology-permuted PPI-GCN representations.

Supplementary Methods 2. Protein representations and graph propagation

Protein node features were generated with the Facebook ESM-2 t33 650M UR50D checkpoint. Residue-level embeddings were mean-pooled to obtain one 1,280-dimensional vector per protein. Proteins longer than 1,022 amino acids were split into chunks and combined by length-weighted averaging. The final feature table covered all 19,622 STRING graph proteins, with no missing graph proteins.

The primary representation-propagation graph was built from STRING v12 human protein links and contained 19,622 nodes and 13,715,404 directed edge-index entries in the retained model input object before adding self-loops. PPI-GCN representations were generated using fixed two-hop GCN-style propagation rather than an end-to-end trainable graph neural network. This design intentionally isolated topology-induced smoothing from capacity gains due to trainable graph layers, so the propagated features were treated as a controlled representation rather than

an optimized end-to-end GNN predictor. Propagation used a binary adjacency derived from the retained edge list; STRING combined scores were not used as propagation weights. The retained edge list was symmetrized, self-loops were added and duplicate edges were coalesced. Let A denote this binary adjacency after symmetrization and self-loop addition, D its degree matrix and X the raw ESM feature matrix. We used symmetric normalization,

$$\hat{A} = D^{-1/2} A D^{-1/2},$$

and formed the propagated representation as

$$H = \frac{1}{3}\left(X + \hat{A}X + \hat{A}^2 X\right).$$

This operation produced one propagated representation per STRING node and did not use variant labels during propagation.

GraphSAGE-style and GAT-style propagated representations were generated as supplementary architecture controls. These controls were included to test whether the representation-level comparison was specific to the fixed PPI-GCN propagation variant. They are treated as architecture controls rather than as an architecture search, because the main contribution of the study is an audit of graph-augmentation gains rather than an optimized predictor design.

For topology profiling, we used a separate thresholded STRING graph based on combined score. The primary topology analyses used a combined-score threshold of 400, yielding 19,622 nodes and 929,472 graph edges. Degree, k-core, community membership, betweenness, clustering coefficient and community-boundary role were computed on this thresholded graph. The representation-propagation graph and the topology-description graph were related STRING-derived objects but not identical analysis objects.

For topology-permutation controls, the graph edges were relabelled by a random node permutation (`edge_index_perm = perm[edge_index]`), while ESM features, labels and train/validation/test variant assignments remained tied to the original protein identities. Variant features were then extracted from the propagated representation of each variant's original mapped STRING protein after propagation on the permuted graph. The graph edge count and unlabeled topology were preserved, but the original protein-to-neighbourhood correspondence was broken. Three graph permutations were combined with three downstream classifier seeds, and the downstream MLP was retrained for each permutation/seed pair.

Supplementary Methods 3. Evaluation splits, homology-aware stress test and statistical analysis

The original evaluation split was constructed at the STRING protein level using seed 42 and an approximate 8:1:1 train, validation and test ratio. Variants assigned to the same STRING protein were kept in the same partition. This produced 141,864 training variants, 17,129 validation variants and 18,220 test variants, with no shared proteins across partitions. The corresponding numbers of unique proteins were 11,298, 1,412 and 1,413 in the train, validation and test sets. All proteins remained embedded in the same full PPI graph, so train and test proteins could still be connected through PPI edges.

For homology-aware evaluation, STRING v12 human protein sequences were clustered with MMseqs using minimum sequence identity 0.40, coverage threshold 0.50 and cov-mode 0. Homologous clusters rather than individual proteins were assigned to train, validation and test partitions, with no protein or homology-cluster overlap across partitions. This split was used as a stricter stress test for reduced sequence-related sharing across partitions, not as a prevalence-matched replacement for the original protein-level split. It preserved the original row counts: 141,864 training variants, 17,129 validation variants and 18,220 test variants. The homology-aware split contained 7,370, 3,315 and 3,438 unique proteins in the train, validation and test sets, respectively, and was built from 9,731 protein clusters with mean cluster size 1.45 and maximum cluster size 251. Because test prevalence changed from 31.95% in the original split to 10.56% in the homology-aware split, AUPRC values should be interpreted within each split rather than directly compared across splits.

Variant-level pathogenicity was predicted with validation-selected multilayer perceptron classifiers. For representation-based and fusion models, each variant inherited the representation of its mapped STRING protein, and the classifier used two hidden layers with 512 and 128 units, ReLU activations, dropout 0.3 and a single-logit output. Low-dimensional external-score-only baselines used a smaller 32/16 classifier with dropout 0.2, while architecture-matched CADD+AlphaMissense and fusion-control comparisons used the 512/128 classifier to

keep downstream capacity matched. Models were trained with BCEWithLogitsLoss and positive-class weighting, using Adam with learning rate 1e-3, weight decay 1e-5, batch size 1024, a maximum of 50 epochs and early stopping after five consecutive validation checks without validation AUC improvement. The positive-class weight was computed from the training partition as `N_negative / N_positive`. The best epoch was selected by validation AUC.

In the evaluated MLP configurations, no additional z-score standardization was applied to CADD, AlphaMissense or embedding-derived inputs. Missing CADD and AlphaMissense values were imputed using training-set medians, and binary missingness indicators were retained as binary indicators. The same preprocessing and feature construction were used across raw ESM, edge-ablated PPI-GCN, topology-permuted PPI-GCN and real PPI-GCN fusion controls. Repeated classifier comparisons used seeds 42, 43 and 44; topology-permuted fusion combined three sampled graph permutations with these classifier seeds. The relative fusion-control comparisons share downstream classifier capacity and preprocessing, whereas absolute fusion performance may depend on feature-scaling choices that were not separately optimized.

Neighbour-risk, neighbour pathogenic-rate and neighbour training-variant-count features were computed from training-set labels only. Validation and test labels were not used to construct these topology-context features. When a test protein had multiple labelled training-neighbour variants, the neighbour pathogenic rate was variant-weighted rather than protein-weighted, so proteins with many labelled neighbouring variants contributed proportionally to the local training-label context.

Uncertainty was evaluated using 1,000 variant-level bootstrap resamples and protein-clustered bootstrap resampling. Variant-level bootstrap resampled individual test variants. Protein-clustered bootstrap resampled test proteins as clusters and retained all variants belonging to each sampled protein, thereby accounting for repeated variants from the same protein. Confidence intervals were reported as empirical 95% bootstrap intervals. Feature-level corrected-sample comparisons used Benjamini-Hochberg false-discovery-rate correction when multiple topology features were tested.

Supplementary Methods 4. Negative-control analysis

Exploratory controls included random raw features, random-feature graph propagation, label shuffling and feature-shuffled GCN propagation. Only the re-materialized feature-shuffled PPI-GCN control is formally reported here because it was retained with complete final-audit provenance. It was rerun using the same downstream evaluation protocol and achieved validation AUC 0.7080, validation AUPRC 0.4241, test AUC 0.5868 and test AUPRC 0.3707. The result is used only as supplementary negative-control evidence and is not part of the main comparative claims.

Supplementary Tables

Supplementary Table 1. Dataset, split and graph summary

Item	Train	Validation	Test	Total or value	Notes
Orig. variants	141,864	17,129	18,220	177,213	Protein split
Orig. benign	102,818	12,737	12,398	127,953	Label 0
Orig. pathogenic	39,046	4,392	5,822	49,260	Label 1
Orig. proteins	11,298	1,412	1,413	14,123 labelled proteins	No protein overlap
Homol. variants	141,864	17,129	18,220	177,213	Cluster split
Homol. benign	96,212	15,445	16,296	127,953	Label 0
Homol. pathogenic	45,652	1,684	1,924	49,260	Label 1
Homol. proteins	7,370	3,315	3,438	14,123 labelled proteins	No protein or cluster overlap

Item	Train	Validation	Test	Total or value	Notes
Homol. clusters	NA	NA	NA	9,731	Mean 1.45; max 251
Repr. graph	NA	NA	NA	19,622 nodes; 13,715,404 directed entries	Propagation graph
Topology graph	NA	NA	NA	19,622 nodes; 929,472 edges	STRING score >=400
CADD matching	NA	NA	NA	177,205 matched; 8 missing	GRCh38 v1.7 whole-genome SNV scores
AlphaMissense matching	NA	NA	NA	171,753 matched; 5,460 missing	hg38 AlphaMissense table

Orig. denotes the original protein-level split. Homol. denotes the homology-aware cluster split. Repr. denotes the representation-propagation graph.

Supplementary Table 2. Representation and architecture-control performance

The original-split raw ESM and PPI-GCN rows report the retained reference run used for paired graph-ablation and bootstrap analyses in Figure 1. Repeated classifier seeds showed the same direction with a smaller mean AUC difference and are reported in the main text.

Split	Model	Val AUC	Val AUPRC	Test AUC	Test AUPRC	Status
Orig.	Raw ESM	0.7814	0.5569	0.7790	0.6074	Reference
Orig.	PPI-GCN	0.8237	0.5821	0.7997	0.6496	Reference
Orig.	Raw ESM (3 seeds)	NA	NA	0.7835 +/- 0.0051	0.6116 +/- 0.0054	3 seeds
Orig.	PPI-GCN (3 seeds)	NA	NA	0.7950 +/- 0.0013	0.6484 +/- 0.0096	3 seeds
Orig.	GraphSAGE-style	0.8008	0.6136	0.7950	0.6044	Arch. control
Orig.	GAT-style	0.8005	0.5876	0.7949	0.6157	Arch. control
Orig.	Degree-only	0.7298	0.4228	0.6377	0.3901	Control
Orig.	Feature-shuffled GCN	0.7080	0.4241	0.5868	0.3707	Rerun verified
Homol.	Raw ESM	0.7315	0.2171	0.7248	0.2418	Stress test
Homol.	PPI-GCN	0.7645	0.2450	0.7606	0.2768	Stress test

Orig. denotes the original split. Homol. denotes the homology-aware split. Arch. denotes architecture.

Supplementary Table 3. External predictor and fusion-control performance

The main fusion-control comparison uses repeated classifier runs where available. Older single-tool rows are retained only as reference context and are not mixed into the matched Figure 2 control panel.

Supplementary Table 3A. Original split

Model	n	Val AUC	Val AUPRC	Test AUC	Test AUPRC
CADD	1	0.9237	0.7855	0.9207	0.8049
AM	1	0.9597	0.9038	0.9470	0.8899

Model	n	Val AUC	Val AUPRC	Test AUC	Test AUPRC
CADD+AM (arch.)	3	0.9667	0.9126	0.9572 +/- 0.00005	0.9068 +/- 0.0003
+ raw ESM	3	0.9686	0.9160	0.9680 +/- 0.0007	0.9317 +/- 0.0029
+ edge-ablated GCN	3	0.9728	0.9257	0.9708 +/- 0.0008	0.9395 +/- 0.0007
+ permuted GCN	9	0.9718	0.9232	0.9694 +/- 0.0008	0.9355 +/- 0.0015
+ real PPI-GCN	3	0.9732	0.9253	0.9718 +/- 0.0007	0.9415 +/- 0.0011

Notes for Table 3A: AM denotes AlphaMissense. "Arch." denotes the architecture-matched CADD+AlphaMissense baseline using the same 512/128 MLP capacity as the fusion controls. The rows beginning with "+" add the listed protein representation to the CADD+AM feature set. Compared with the architecture-matched CADD+AM baseline, raw ESM fusion increased test AUC/AUPRC by +0.0108/+0.0249, edge-ablated PPI-GCN fusion by +0.0136/+0.0327, topology-permuted PPI-GCN fusion by +0.0122/+0.0286 and real PPI-GCN fusion by +0.0146/+0.0347. The edge-ablated row uses one deterministic direct train-test edge-ablation graph evaluated across three classifier seeds, whereas the topology-permuted row combines three random graph permutations with three classifier seeds.

Supplementary Table 3B. Homology-aware split

Model	n	Val AUC	Val AUPRC	Test AUC	Test AUPRC
CADD	1	0.9506	0.6713	0.9504	0.6895
AM	1	0.9614	0.7945	0.9632	0.8029
CADD+AM (arch.)	3	0.9739	0.8362	0.9754 +/- 0.00005	0.8401 +/- 0.0002
+ raw ESM	3	0.9715	0.8228	0.9728 +/- 0.0003	0.8249 +/- 0.0016
+ PPI-GCN	1	0.9732	0.8273	0.9741	0.8333

Notes for Table 3B: the homology-aware graph-fusion row is a single retained point estimate. It is reported as a stress-test result, not as a fully symmetric repeated-ensemble comparison. Relative to the repeated-run architecture-matched CADD+AlphaMissense mean, the retained graph-fusion point estimate was -0.0013 AUC and -0.0068 AUPRC.

Supplementary Table 4. Uncertainty and repeated-run summary

Supplementary Table 4A. Raw ESM versus PPI-GCN uncertainty

Bootstrap	Metric	Delta	CI low	CI high	n
Variant	AUC	+0.0208	+0.0174	+0.0240	1,000
Variant	AUPRC	+0.0420	+0.0353	+0.0484	1,000
Protein-clustered	AUC	+0.0207	-0.0080	+0.0527	1,000
Protein-clustered	AUPRC	+0.0421	-0.0021	+0.0798	1,000

Supplementary Table 4B. Original-split strong-predictor fusion-control uncertainty Comparison codes: C0, real PPI-GCN fusion versus architecture-matched CADD+AlphaMissense; C1, real PPI-GCN fusion versus raw ESM fusion; C2, real PPI-GCN fusion versus edge-ablated PPI-GCN fusion; C3, real PPI-GCN fusion versus topology-permuted PPI-GCN fusion.

ID	Metric	Delta	CI low	CI high	n
C0	AUC	+0.0143	+0.0062	+0.0224	1,000
C0	AUPRC	+0.0341	+0.0167	+0.0566	1,000
C1	AUC	+0.0038	+0.0005	+0.0075	1,000
C1	AUPRC	+0.0093	+0.0032	+0.0167	1,000
C2	AUC	+0.0009	-0.0009	+0.0027	1,000
C2	AUPRC	+0.0017	-0.0018	+0.0052	1,000
C3	AUC	+0.0019	-0.0005	+0.0052	1,000
C3	AUPRC	+0.0050	-0.0002	+0.0112	1,000

Supplementary Table 4C. Earlier retained-reference fusion bootstrap retained as context Comparison code: R0, earlier retained-reference graph fusion versus earlier retained-reference CADD+AlphaMissense; not architecture-matched.

ID	Metric	Delta	CI low	CI high	n
R0	AUC	+0.0133	+0.0119	+0.0146	1,000
R0	AUPRC	+0.0310	+0.0268	+0.0350	1,000

Table 4A shows that variant-level bootstrap supports positive retained-reference PPI-GCN gains, whereas protein-clustered bootstrap widens uncertainty. Table 4B shows that the real PPI-GCN fusion margin over raw ESM fusion remained positive under protein-clustered paired bootstrap, but the smaller margins over edge-ablated and topology-permuted graph controls crossed zero. Table 4C is retained as an earlier retained-reference context analysis and is not architecture-matched.

The homology-aware strong-predictor bootstrap row comes from an earlier retained reference-run comparison rather than the architecture-matched repeated ensemble. The main manuscript reports this result as a stress-test pattern with a retained graph-fusion point estimate.

Supplementary Table 5. Topology-stratified PPI-GCN gain over raw ESM

Stratum	n	Positive rate	Delta AUC	Delta AUPRC
Degree-low	2,795	0.0426	+0.0314	+0.0109
Degree-mid	4,762	0.2089	+0.0339	+0.0280
Degree-high	10,663	0.4415	-0.0178	-0.0021
K-core-low	2,847	0.0425	+0.0297	+0.0116
K-core-mid	5,685	0.3078	+0.0775	+0.1049
K-core-high	9,688	0.4078	-0.0276	+0.0015
Risk-low	6,323	0.1056	+0.0641	+0.0631
Risk-mid	5,829	0.2882	+0.0265	+0.0515
Risk-high	6,019	0.5772	+0.0284	+0.0270
Betweenness-low	3,050	0.1393	+0.0614	+0.0769
Betweenness-mid	4,486	0.1899	+0.0522	+0.0501
Betweenness-high	10,684	0.4254	-0.0145	+0.0027
Clustering-low	8,234	0.2888	-0.0036	-0.0099
Clustering-mid	5,950	0.4198	+0.0544	+0.0444
Clustering-high	4,036	0.2344	+0.0348	+0.0941
Boundary-like	6,198	0.3527	+0.0182	+0.0200
Boundary-mixed	6,598	0.2963	+0.0140	+0.0503
Boundary-internal	5,397	0.3111	+0.0383	+0.0151

Selected exploratory joint topology cells:

Cell codes: J0, mid k-core / high betweenness; J1, mid k-core / mid clustering; J2, high neighbour risk / mid k-core; J3, high k-core / low clustering.

ID	n	Pos. rate	Delta AUC	Delta AUPRC	AUC CI
J0	1,950	0.4297	+0.1429	+0.1533	+0.1246 to +0.1617
J1	2,089	0.4045	+0.1408	+0.1428	+0.1234 to +0.1581
J2	1,904	0.5767	+0.1274	+0.0983	+0.1045 to +0.1499
J3	4,791	0.3870	-0.0532	-0.0480	-0.0611 to -0.0457

Full topology maps are provided in the prepared source-data archive rather than printed in full in the PDF supplement.

These joint cells were selected from the same test-set topology map used for estimation. Their bootstrap intervals describe within-cell sampling uncertainty without correcting for the selection step. The cells are exploratory gain profiles, not independently validated topology-defined subgroups.

Supplementary Table 6. Corrected-sample and difficulty-matched statistics

Corrected-sample group counts for raw ESM to PPI-GCN:

All strong-predictor transition counts in this table correspond to the exploratory retained-reference model pair used in Figure 4, not to the architecture-matched repeated ensembles used in Figure 2.

Group	n	Positive rate
both correct	11,875	0.3270
both wrong	3,965	0.2678
improved/corrected	1,436	0.5829
degraded	944	0.0424
false negatives rescued	837	1.0000
false positives suppressed	599	0.0000

Corrected-sample group counts for retained-reference CADD+AlphaMissense to retained-reference graph fusion:

Group	n	Positive rate
both correct	15,981	0.3173
both wrong	1,298	0.3274
improved/corrected	666	0.3333
degraded	275	0.3818
false negatives rescued	222	1.0000
false positives suppressed	444	0.0000

Selected false-positive-suppressed versus false-negative-rescued topology contrasts for raw ESM to PPI-GCN:

Feature	FP mean	FN mean	Diff	FDR
degree bucket	0.8698	1.9833	-1.1135	4.48e-188
k-core bucket	0.8381	1.1254	-0.2874	3.12e-22
neighbour risk bucket	0.5726	1.6941	-1.1215	2.74e-129
neighbour pathogenic rate	0.2717	0.4329	-0.1612	5.95e-90
betweenness bucket	0.9482	1.8530	-0.9048	1.82e-120
clustering coefficient	0.2534	0.2022	+0.0512	1.16e-07
boundary role bucket	1.1519	0.3321	+0.8198	2.12e-87

Selected false-positive-suppressed versus false-negative-rescued topology contrasts for retained-reference CADD+AlphaMissense to retained-reference graph fusion:

Feature	FP mean	FN mean	Diff	FDR
degree bucket	1.5541	1.8198	-0.2658	2.60e-06
k-core bucket	1.5518	1.6036	-0.0518	0.8373
neighbour risk bucket	0.7793	1.7523	-0.9730	4.34e-55
neighbour pathogenic rate	0.3276	0.4955	-0.1679	1.57e-56
betweenness bucket	1.5541	1.7252	-0.1712	2.96e-04
clustering coefficient	0.1807	0.2176	-0.0368	1.64e-14
boundary role bucket	0.9685	0.7207	+0.2477	3.81e-05

Difficulty-matched rescued false-negative contrasts:

Feature	Corrected subset mean	Persistent error mean	Hedges' g	FDR
Degree bucket	1.9833	1.5546	+0.9675	0.0011
K-core bucket	1.1254	1.5358	-0.8441	0.0011
Neighbour risk bucket	1.6941	1.0904	+0.8163	0.0011
Neighbour pathogenic rate	0.4329	0.3793	+0.5259	0.0011
Betweenness bucket	1.8530	1.5348	+0.5923	0.0011
Clustering coefficient	0.2022	0.2322	-0.4011	0.0011
Boundary role bucket	0.3321	0.9953	-0.8000	0.0011
Within-community neighbour fraction	0.5655	0.7093	-0.7950	0.0011

These association and stratification results do not establish a causal mechanism.

Supplementary Table 7. Negative-control audit

The formal supplementary table reports only the negative-control result that was rerun and re-materialized with complete final-audit provenance. Older random-feature and label-shuffling summaries are not used as formal evidence in this submission.

Control	Split	n	Val A	Val AP	Test A	Test AP
Feat.-shuffled GCN	Orig.	1	0.7080	0.4241	0.5868	0.3707

Val A and Test A denote validation and test AUC, respectively. Val AP and Test AP denote validation and test AUPRC. This control is used only as supplementary negative-control evidence and is not part of the main comparative claims.

Source-data archive status

The prepared compact source-data archive contains figure source data for Figures 1-4, topology-stratified metrics, joint topology maps, corrected-sample statistics, fusion-control summaries, bootstrap outputs and checksum files. Large third-party resources, including provider-controlled CADD and AlphaMissense score files, are not redistributed and should be obtained from their original providers.